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CODE:

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# Perceptron Based Iris Classification

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import Perceptron
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# Load Iris Dataset
iris = load_iris()

# Input features
X = iris.data

# Target classes
y = iris.target

# Split the dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

# Scale the data
scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Create Perceptron model
model = Perceptron(
    max_iter=1000,
    eta0=0.1,
    random_state=42
)

# Train the model
model.fit(X_train_scaled, y_train)

# Predict the test data
y_pred = model.predict(X_test_scaled)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)

# Display results
```

```
print("PERCEPTRON BASED IRIS CLASSIFICATION")
print("=" * 50)

print("\nAccuracy:", round(accuracy * 100, 2), "%")

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

print("\nClassification Report:")
print(classification_report(
    y_test,
    y_pred,
    target_names=iris.target_names,
    zero_division=0
))

# Classify a new Iris flower
new_flower = [[5.1, 3.5, 1.4, 0.2]]

# Scale the new sample
new_flower_scaled = scaler.transform(new_flower)

# Predict the flower class
prediction = model.predict(new_flower_scaled)

print("\nNEW FLOWER DETAILS")
print("-" * 50)
print("Sepal Length:", new_flower[0][0])
print("Sepal Width :", new_flower[0][1])
print("Petal Length:", new_flower[0][2])
print("Petal Width :", new_flower[0][3])

print("\nPredicted Flower:",
      iris.target_names[prediction[0]])
```

OUTPUT :

```
PERCEPTRON BASED IRIS CLASSIFICATION
==================================================

Accuracy: 86.67 %

Confusion Matrix:
[[10  0  0]
 [ 0  7  3]
 [ 0  1  9]]

Classification Report:
              precision    recall  f1-score   support

      setosa       1.00      1.00      1.00        10
  versicolor       0.88      0.70      0.78        10
   virginica       0.75      0.90      0.82        10

    accuracy                           0.87        30
   macro avg       0.88      0.87      0.87        30
weighted avg       0.88      0.87      0.87        30


NEW FLOWER DETAILS
--------------------------------------------------
Sepal Length: 5.1
Sepal Width : 3.5
Petal Length: 1.4
Petal Width : 0.2

Predicted Flower: setosa
```